

Task 1:

TASK 1: Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency

Code:

```
gopinath@GOPINATH: ~
GNU nano 8.7.1

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>
#include <string.h>

#define DATA_SIZE 100000

int main()
{
    int pipefd[2];
    pid_t pid;
    char buffer[1024];
    char data[] = "Producer sends data to Consumer through pipe.\n";
    long total_bytes = 0;

    struct timespec start, end;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        exit(1);
    }

    if (pid == 0)
    {
        // Child - Consumer
        close(pipefd[1]);

        while (1)
        {
            int n = read(pipefd[0], buffer, sizeof(buffer));

            if (n <= 0)
                break;

            total_bytes += n;

            close(pipefd[0]);

            printf("Consumer received %ld bytes.\n", total_bytes);
            exit(0);
        }
    }
    else
    {
        // Parent - Producer
        close(pipefd[0]);

        clock_gettime(CLOCK_MONOTONIC, &start);

        for (int i = 0; i < DATA_SIZE; i++)
        {
            write(pipefd[1], data, strlen(data));
        }

        close(pipefd[1]);

        wait(NULL);

        clock_gettime(CLOCK_MONOTONIC, &end);

        double time_taken =
            (end.tv_sec - start.tv_sec) +
            (end.tv_nsec - start.tv_nsec) / 1e9;

        long bytes_sent = DATA_SIZE * strlen(data);

        printf("Producer sent %ld bytes.\n", bytes_sent);
        printf("Communication time: %.6f seconds\n", time_taken);

        if (time_taken > 0)
        {
            double throughput =
                (bytes_sent / time_taken) / (1024 * 1024);

            printf("Communication throughput: %.2f MB/s\n",
                throughput);
        }
    }

    return 0;
}
```

Output:

gopinath@GOPINATH

```
gopinath@GOPINATH:~$ gcc w5.c -o w5
gopinath@GOPINATH:~$ ./w5
Consumer received 4600000 bytes.
Producer sent 4600000 bytes.
Communication time: 0.147341 seconds
Communication throughput: 29.77 MB/s
gopinath@GOPINATH:~$
```

Observation:

The **parent process produces data and writes it into the anonymous pipe**, while the **child process reads and consumes the data**. Communication efficiency is measured using the **time taken and data throughput (MB/s)**.

Task 2:

Develop a program that executes the equivalent of the shell command: `ls -l | grep` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code:

```
gopinath@GOPINATH: ~
GNU nano 8.7.1

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    // Create pipe
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    // Create first child for ls -l
    pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        // Redirect stdout to pipe
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        // Execute ls
        execlp("ls", "ls", "-l", (char *)NULL);

        perror("execlp ls");
        exit(1);
    }

    // Create second child for grep
    pid2 = fork();

    if (pid2 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid2 == 0)
    {
        // Redirect stdin from pipe
        dup2(pipefd[0], STDIN_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        // Execute grep
        execlp("grep", "grep", ".", (char *)NULL);

        perror("execlp grep");
        exit(1);
    }

    // Parent closes both ends of pipe
    close(pipefd[0]);
    close(pipefd[1]);

    // Wait for both children
    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);

    printf("Parent: Both processes completed.\n");

    return 0;
}
```

Output:

```
gopinath@GOPINATH
gopinath@GOPINATH:~$ gcc w51.c
gopinath@GOPINATH:~$ ./a.out
total 268
-rw-r--r-- 1 gopinath gopinath  57 Aug  6 14:47 First.c
drwxr-xr-x 3 gopinath gopinath 4096 Aug  3 05:02 Hello-World
drwxr-xr-x 11 gopinath gopinath 4096 Jul 30 09:39 OSSP
-rw-r--r-- 1 gopinath gopinath 1322 Aug 25 03:28 P-week2.c
-rw-r--r-- 1 gopinath gopinath 1206 Aug  7 15:18 Practical2.c
-rw-r--r-- 1 gopinath gopinath  935 Aug  6 15:00 Skill12.c
-rw-r--r-- 1 gopinath gopinath  654 Aug  7 14:32 Skill13.c
-rw-r--r-- 1 gopinath gopinath  802 Aug 14 14:48 Skill13Task2.c
-rwxr-xr-x 1 gopinath gopinath 16336 Aug 27 09:50 a.out
-rw-r--r-- 1 gopinath gopinath  936 Aug  5 16:22 command_exec.c
-rw-r--r-- 1 gopinath gopinath    0 Aug 25 04:47 copy.txt
-rw-r--r-- 1 gopinath gopinath   24 Aug 25 04:56 destination.txt
```

```

-rw-r--r-- 1 gopinath gopinath    1 Jul 28 14:48 nano.565.save
-rw-r--r-- 1 gopinath gopinath   729 Aug 25 04:43 new.c
-rw-r--r-- 1 gopinath gopinath   270 Aug 25 04:46 new1.c
-rwxr-xr-x 1 gopinath gopinath 16216 Aug 25 04:56 new12
-rw-r--r-- 1 gopinath gopinath  1252 Aug 25 04:55 new12.c
-rw-r--r-- 1 gopinath gopinath    710 Aug 20 08:27 os-p-2.c
-rw-r--r-- 1 gopinath gopinath    931 Aug 20 08:27 os.c
-rw-r--r-- 1 gopinath gopinath    156 Aug 18 07:00 os1
drwxr-xr-x 6 gopinath gopinath  4096 Aug  4 08:12 ossp
-rw-r--r-- 1 gopinath gopinath    148 Aug 19 09:50 s1.c
-rw-r--r-- 1 gopinath gopinath   881 Aug 19 23:47 s12.c
-rw-r--r-- 1 gopinath gopinath   559 Aug 19 23:57 s2.c
-rw-r--r-- 1 gopinath gopinath   538 Aug 20 00:00 s21.c
-rw-r--r-- 1 gopinath gopinath     25 Aug 25 05:01 sample.txt
-rw-r--r-- 1 gopinath gopinath   3735 Aug 22 06:33 sk1.c
-rw-r--r-- 1 gopinath gopinath  1905 Aug 11 07:08 skil4Task1.c
-rw-r--r-- 1 gopinath gopinath    968 Aug  7 14:52 skill13Task2.c
-rw-r--r-- 1 gopinath gopinath   5428 Aug 20 06:36 skillweek2.c
-rw-r--r-- 1 gopinath gopinath   1287 Aug 20 09:05 skillweek4.c
-rw-r--r-- 1 gopinath gopinath   4192 Aug 20 06:39 skilweek2.c
-rw-r--r-- 1 gopinath gopinath   5507 Aug 20 06:58 skilweek21.c
-rw-r--r-- 1 gopinath gopinath     29 Aug 25 05:00 source.txt
-rw-r--r-- 1 gopinath gopinath   1885 Aug 20 05:28 sweek2.c
-rw-r--r-- 1 gopinath gopinath   1855 Aug 20 05:34 sweek2.c.save
-rw-r--r-- 1 gopinath gopinath   1322 Aug 25 03:29 w-p2.c
-rwxr-xr-x 1 gopinath gopinath 16424 Aug 27 09:45 w5
-rw-r--r-- 1 gopinath gopinath   1771 Aug 27 09:42 w5.c
-rwxr-xr-x 1 gopinath gopinath 16336 Aug 27 09:49 w51
-rw-r--r-- 1 gopinath gopinath   1305 Aug 27 09:48 w51.c
-rw-r--r-- 1 gopinath gopinath   3010 Aug 24 14:05 wTask42.c
-rw-r--r-- 1 gopinath gopinath   1472 Aug 22 06:12 weeek4.c
-rw-r--r-- 1 gopinath gopinath   2926 Aug 24 13:41 weeekTask4.c
-rw-r--r-- 1 gopinath gopinath    920 Aug 27 08:09 week4.c
-rwxr-xr-x 1 gopinath gopinath 16256 Aug 27 08:37 week42
-rw-r--r-- 1 gopinath gopinath    811 Aug 27 08:19 week42.c
-rw-r--r-- 1 gopinath gopinath   2688 Aug 22 06:19 week4Task1.c
-rw-r--r-- 1 gopinath gopinath    926 Aug 27 08:14 wek41.c
Parent: Both processes completed.
gopinath@GOPINATH:~$ █

```

Observation:

- The program successfully implements the **ls -l | grep .** command using **fork()**, **pipe()**, **dup2()**, and **exec()**.
- It demonstrates **inter-process communication**, where the output of **ls** is passed through the pipe to **grep**.